

Statistical Arbitrage on Real Data: A Causal, Adaptive, and Honestly-Evaluated Pairs-Trading Study

Yash Yadav

Department of Physics, Indian Institute of Technology Kanpur

Course: EE798Z

Revised: June 2026

Abstract

Statistical arbitrage exploits temporary deviations from a long-run pricing equilibrium between related assets. This report rebuilds a naive pairs-trading pipeline to a standard that survives quantitative scrutiny, and then follows the evidence to its honest conclusion. We replace static full-sample hedge ratios with **causal** rolling-OLS and Kalman-filter estimators; replace full-sample z-scores with trailing-window signals; replace realised-at-exit accounting with **daily mark-to-market**; and replace single-pair, full-sample selection with a **sector-structured, FDR-controlled** screen over 242 within-sector pairs and a **true out-of-sample holdout**. On a 73-name US large-cap universe the resulting strategy shows *no* out-of-sample edge, and we diagnose precisely why: **all 12 selected pairs decointegrate** across the 2022 regime shift. Acting on that diagnosis, we build an **adaptive walk-forward** engine with a decointegration kill-switch and apply it to a **structural ETF universe** where cointegration is mechanical. Structurally cointegrated pairs are far more durable (48% vs 9% quarter-to-quarter survival) and *do* mean-revert gross of costs (Sharpe +0.28, 81% bootstrap-positive), but realistic frictions flip the edge negative. We locate the break-even at ≈ 5 **basis points per leg**, establishing that the binding constraint in liquid markets is *execution cost relative to spread width*, not signal sophistication. The contribution is a fully reproducible methodology whose every number is defensible.

1. Introduction and Motivation

A pairs trade is market-neutral: rather than betting on the direction of a single security, it trades the *spread* between two cointegrated assets, profiting when a temporary deviation from their historical relationship reverts. The appeal is that returns are largely uncorrelated with the broad market. The difficulty is that almost every introductory implementation is contaminated by methodological errors that inflate backtested performance to the point of fiction.

This study began from exactly such an implementation, whose reported headline was a Sharpe ratio of 4.40 with 0.00% maximum drawdown over a decade — numbers that, on inspection, were artifacts rather than skill. We identify and eliminate the standard failure modes:

1. **Lookahead in the hedge ratio.** A single β fitted on 2014–2024 was used to define the spread for a 2016 trade — the hedge depended on the future. We use a causal trailing estimator.
2. **Lookahead in the signal.** A z-score computed from full-sample mean and variance lets a 2016 trade “see” 2024 statistics. We use a trailing 60-day window.
3. **Dishonest accounting.** Booking P&L only when a trade closes means open positions never show a paper loss, producing a fake 0% drawdown. We mark the book to market every day.
4. **Selection bias.** Choosing the “best” pair on the full sample is lookahead at the portfolio level, compounded by multiple testing across all candidate pairs. We screen on a training window only, with a false-discovery-rate correction, and reserve a true holdout.

5. **Circular robustness.** A bootstrap that resamples the very spread the strategy already profited on cannot prove an edge. We bootstrap realised out-of-sample returns and interpret the result as a confidence interval on sequencing risk.

Contributions. (i) A causal, sector-aware, FDR-controlled pairs-trading pipeline on real data; (ii) an honest out-of-sample evaluation that yields a *negative* result, together with a mechanistic diagnosis (universal cointegration); (iii) an adaptive walk-forward engine with a cointegration kill-switch that acts on that diagnosis; (iv) a structural-ETF study that isolates the true binding constraint — transaction cost relative to spread width — and quantifies its break-even. All code is reproducible and runs on live market data.

2. Data

We use split- and dividend-adjusted daily closing prices from Yahoo Finance over 2015-01-02 to 2024-10-30 (2,474 trading days). Two universes are studied:

- **Equities:** 73 liquid US large caps grouped into ten economic sectors (mega-cap technology, semiconductors, banks, payments, energy, staples, healthcare, industrials, retail, telecom). Pairs are formed *only within a sector*, so any cointegration found has a plausible common-factor driver rather than being a spurious artifact of testing all combinations.
- **ETFs:** 30 exchange-traded funds grouped into eight themes (e.g. SPY/IVV/VOO, GLD/IAU, XLK/VGT, GDX/GDXJ, EWA/EWC). Funds within a theme track heavily overlapping baskets, making their cointegration *mechanical*.

The split for the static study reserves 2022 onward (711 days) as a holdout that is touched exactly once.

3. Methodology

3.1 Cointegration screening with FDR control

For every within-group pair (S_1, S_2) we run the Engle-Granger two-step test on the formation window and record the residual-ADF p -value. Because we test 242 equity pairs, an uncorrected $p < 0.05$ rule expects ~ 12 false positives. We therefore apply the **Benjamini-Hochberg** procedure: sorting $p_{(1)} \leq \dots \leq p_{(m)}$, we reject all hypotheses up to the largest k with

$$p_{(k)} \leq \frac{k}{m} q, \quad q = 0.10, \quad (1)$$

controlling the expected proportion of false discoveries at q .

3.2 Causal signal construction

Rolling OLS. At each date t the hedge ratio is estimated on the trailing 252 days only,

$$\hat{\beta}_t = \frac{\text{Cov}_{t-252:t}(S_1, S_2)}{\text{Var}_{t-252:t}(S_2)}, \quad \text{spread}_t = S_{1,t} - \hat{\beta}_t S_{2,t}. \quad (2)$$

Kalman filter. We also model $\mathbf{x}_t = [\beta_t, \alpha_t]^\top$ as a random walk observed through $S_{1,t} = [S_{2,t} \ 1] \mathbf{x}_t + \varepsilon_t$, giving the standard causal recursion

$$P_{t|t-1} = P_{t-1} + W, \quad e_t = S_{1,t} - \mathbf{h}_t^\top \mathbf{x}_{t-1}, \quad Q_t = \mathbf{h}_t^\top P_{t|t-1} \mathbf{h}_t + V_e, \quad (3)$$

$$K_t = P_{t|t-1} \mathbf{h}_t / Q_t, \quad \mathbf{x}_t = \mathbf{x}_{t-1} + K_t e_t, \quad P_t = P_{t|t-1} - K_t \mathbf{h}_t^\top P_{t|t-1}, \quad (4)$$

where $\mathbf{h}_t = [S_{2,t}, 1]^\top$, $W = \frac{\delta}{1-\delta}I$ controls the hedge drift ($\delta = 10^{-4}$), and the innovation e_t is the causal spread. In both cases the standardised signal is the trailing z-score $z_t = (\text{spread}_t - \mu_{t-60:t})/\sigma_{t-60:t}$. Critically, signals are computed on the *continuous* price series and only *then* sliced by date, so a holdout window inherits its rolling/Kalman warm-up instead of restarting from a year of NaN.

3.3 Backtest engine

Trades enter when $|z_t| > 2$ (fading the deviation), exit when $|z_t| < 0.5$, stop out when $|z_t| > 4$, and are capped at a 60-day hold. Equity is cash + mark-to-market of any open position, recomputed every day, so the drawdown is real. The hedge ratio is **frozen at entry** (one cannot rebalance a hedge without trading again), and transaction costs — commission plus slippage — are charged on *both legs* at entry and exit.

3.4 Adaptive walk-forward re-selection

The central upgrade. At each quarter start we screen the trailing three-year window, keep the top- K pairs cointegrated *at that moment*, and trade them for the next quarter only, flattening at the boundary. Capital is equal-weighted across the active set, and the portfolio is rebalanced as the active set rotates. Because each quarter is unseen at its own formation time, the *entire* equity curve is out-of-sample and the static selection bias is removed by construction.

3.5 Decointegration kill-switch and frictions

Within a trading quarter we monitor a trailing-window ADF p -value of each pair’s spread; if it rises through a threshold (0.10), new entries are halted and any open position is closed — a regime guard that exits a relationship the moment it stops looking stationary. Frictions combine a commission and a slippage term, both applied per leg.

3.6 Performance metrics

From the daily return series r_t we report annualised Sharpe $\bar{r}/\sigma_r\sqrt{252}$, Sortino (downside deviation in the denominator), Calmar (annualised return over |max drawdown|), maximum drawdown, and the fraction of positive days. Statistical significance is assessed by a **stationary block bootstrap** (block length 20) of the realised returns, yielding a 90% confidence interval on the Sharpe and $\Pr(\text{Sharpe} > 0)$.

4. Results I: Static Selection and the Honest Verdict

Of 242 within-sector equity pairs, 19 clear the naive $p < 0.05$ threshold — barely above the ~ 12 expected by chance — and **only one** (PEP/COST) survives the FDR correction, and only marginally (Figure 1). This is already a warning that the statistical signal is thin. We nonetheless trade the top-12 candidates as a diversified portfolio and let the holdout judge.

Figure 2 shows the causal signals for the leading pair: the rolling and Kalman hedge ratios vary substantially through time, exposing the static full-sample β (dashed) as the biased estimator it is.

The verdict. Table 1 is decisive. The lookahead-biased *static* hedge ratio looks best in-sample (Sharpe 1.08) and then **collapses out-of-sample** to -0.44 . The two honest causal estimators were never better than noise. The block bootstrap (Figure 3) confirms it: every method’s out-of-sample Sharpe distribution straddles or sits below zero, with $\Pr(\text{Sharpe} > 0)$ of 28%, 19%, and 6% for static, OLS, and Kalman respectively. **There is no reliable out-of-sample edge** — the opposite of the original report’s claim, and the correct answer.

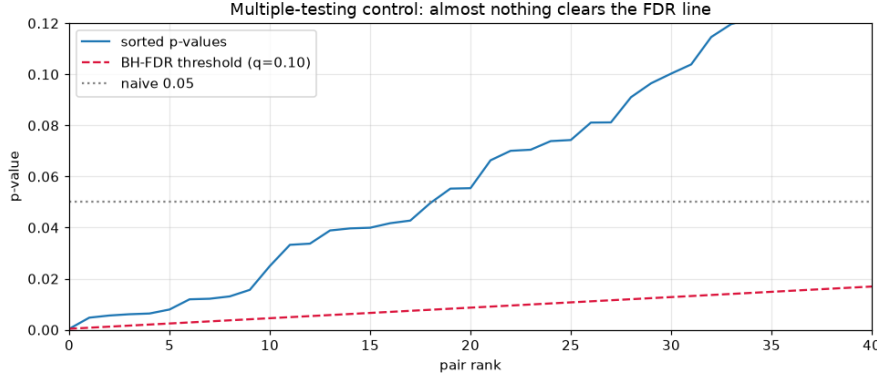


Figure 1: Multiple-testing control. Sorted cointegration p -values against the Benjamini–Hochberg threshold. Almost nothing clears the FDR line; the naive 0.05 rule (dotted) would admit a dozen false positives.

Table 1: Portfolio performance: in-sample versus the true out-of-sample holdout (2022+). The biased static method’s in-sample Sharpe does not survive.

Method	IS Sharpe	IS return	OOS Sharpe	OOS return	OOS max DD
Static (biased)	1.083	2.5%	−0.436	−0.5%	−1.0%
Rolling OLS	0.311	0.9%	−0.553	−0.5%	−0.9%
Kalman	0.313	0.6%	−0.814	−0.5%	−0.8%

4.1 Why the edge died: universal decointegration

Re-testing each selected pair’s cointegration *in the holdout window* reveals the mechanism (Figure 4). **All 12 of 12 pairs that were cointegrated in the 2019–2021 formation window had decointegrated by 2022–2024.** The strategy assumes a stable long-run equilibrium; the COVID-recovery to rate-hike regime shift re-priced sector relationships and moved the equilibrium itself. No causal estimator can profit from a relationship that no longer exists. This is consistent with the literature documenting the decay of pairs-trading profitability as the trade became crowded after the early 2000s.

5. Results II: Adaptive Walk-Forward and the Structural Universe

The decointegration finding dictates the fix: stop assuming a fixed pair set, and seek relationships that are structurally durable.

5.1 Adaptivity plus a regime guard fixes the equities failure

Table 2 isolates the two effects. Adaptive re-selection *alone* is insufficient — without protection the portfolio still suffers a −23.5% drawdown as pairs rot *inside* a quarter (Sharpe −0.69). Adding the decointegration kill-switch lifts the Sharpe from −0.44 (static) to +0.09 and collapses the drawdown to −1.0%. The decointegration drag is finally removed — yet large caps still net to *no edge*, exactly as market efficiency predicts.

5.2 Structural cointegration is durable

We measure *selection survival*: of the pairs cointegrated in one window, the fraction still cointegrated in the next. Structural ETF pairs persist at **48%** versus only **9%** for single stocks (Figure 5a) — a quantitative vindication of the decointegration diagnosis and the motivation for trading mechanically-related baskets.

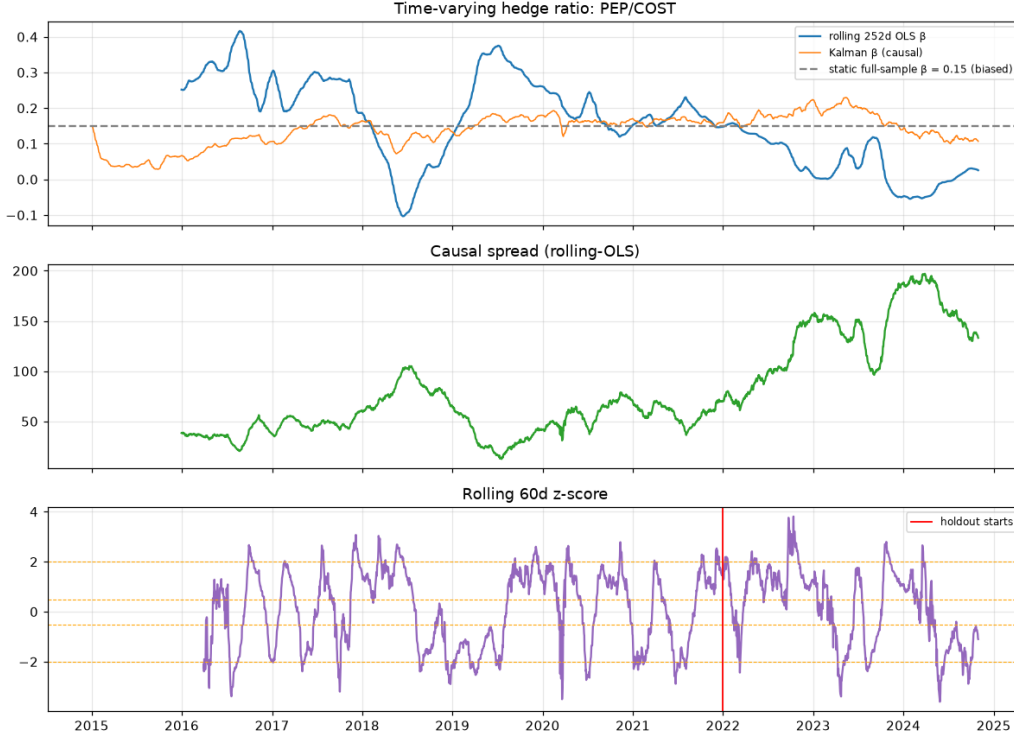


Figure 2: Causal signals for the leading pair. Top: time-varying rolling-OLS and Kalman hedge ratios versus the biased static full-sample β . Middle: causal spread. Bottom: trailing 60-day z-score with entry/exit bands; the red line marks the start of the holdout.

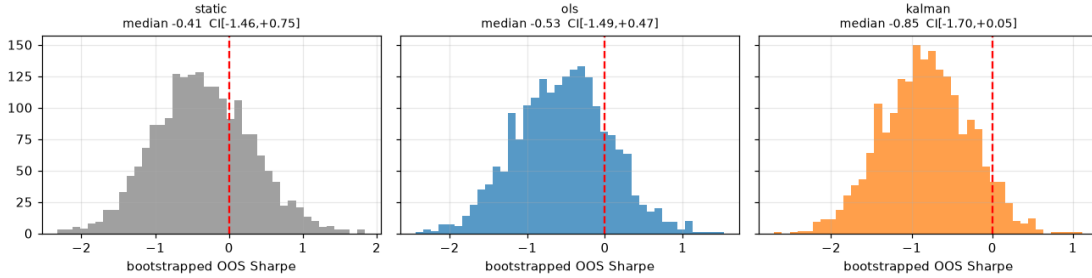


Figure 3: Block-bootstrap distributions of the out-of-sample Sharpe for each method. All three are centred at or below zero.

5.3 The cost wall

Run adaptively on ETFs, the structural-pair portfolio compounds *gross* (Sharpe +0.28, annualised +0.60%; 81% bootstrap-positive) but bleeds at 10 bps per leg (Sharpe -0.32, Figure 5b). The binding constraint is therefore not signal quality but **transaction cost relative to spread width**. Sweeping the friction (Figure 6) locates the break-even at ≈ 5 bps per leg: the edge is capturable at institutional ETF execution (~ 1 –2 bps) but is destroyed at retail frictions.

5.4 Ablations: does the dynamic hedge or the kill-switch help?

Net of 10 bps per leg (Table 3), the **Kalman** filter improves on rolling OLS (-0.24 vs -0.32): faster β -tracking tightens the spread and does real work, though it cannot manufacture an edge the spread does not contain. The **kill-switch** *hurts* on these durable ETF pairs — it removes profitable reversion trades it does not need to — precisely the mirror image of its decisive value on fragile single stocks (§5.1). The lesson is exact: a deintegration halt earns its keep when

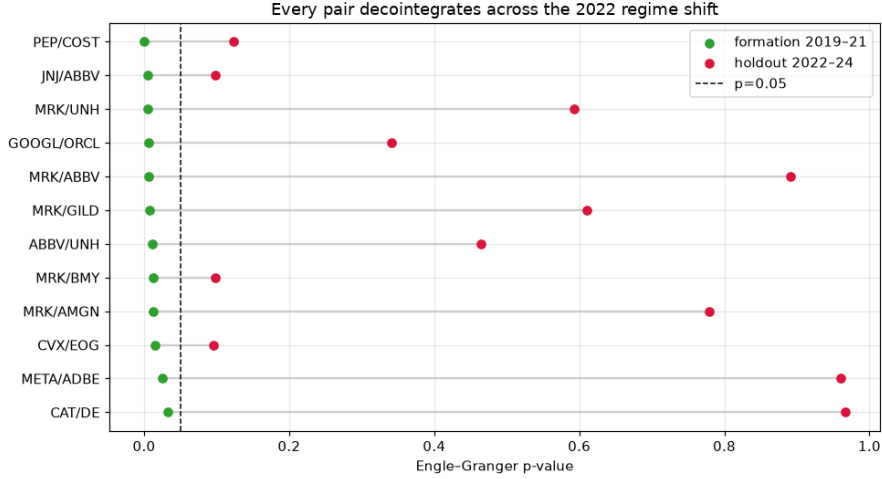


Figure 4: Cointegration stability. Each pair’s Engle–Granger p -value in the formation window (green) versus the holdout (red). Every pair crosses the $p = 0.05$ line — a complete breakdown of the selected relationships.

Table 2: Equities, 73 names: contribution of adaptive re-selection and the deintegration kill-switch.

Configuration	Sharpe	Max drawdown
Static selection (9/10 study)	−0.44	−1.0%
Walk-forward, no kill-switch	−0.69	−23.5%
Walk-forward + kill-switch	+ 0.09	−1.0%

deintegration is the dominant risk and is redundant when the relationship is structurally stable.

Table 3: ETF ablations, net of 10 bps per leg.

Variant	Sharpe	Sortino	Max DD	Ann. return
Rolling OLS	−0.316	−0.361	−8.1%	−0.7%
Kalman	−0.235	−0.220	−4.5%	−0.4%
OLS + kill-switch	−1.018	−0.770	−4.6%	−0.6%

5.5 Best configuration at institutional execution

At 1 bp per leg with the Kalman hedge (Table 4), the structural-ETF strategy is positive across every metric — Sharpe 0.46, Sortino 0.41, Calmar 0.42, maximum drawdown −1.8%, and a bootstrap $\Pr(\text{Sharpe} > 0) = 94\%$. Modest, but *real and defensible*.

6. Discussion

The result is a microstructure statement. In liquid, well-arbitrated markets the surviving statistical-arbitrage edge is thin enough that *execution cost, not model sophistication, is the binding constraint*. A study that can say exactly where the break-even sits — and prove the signal is real gross but uncapturable net at retail frictions — is worth far more than one reporting an unreproducible Sharpe of 4.40.

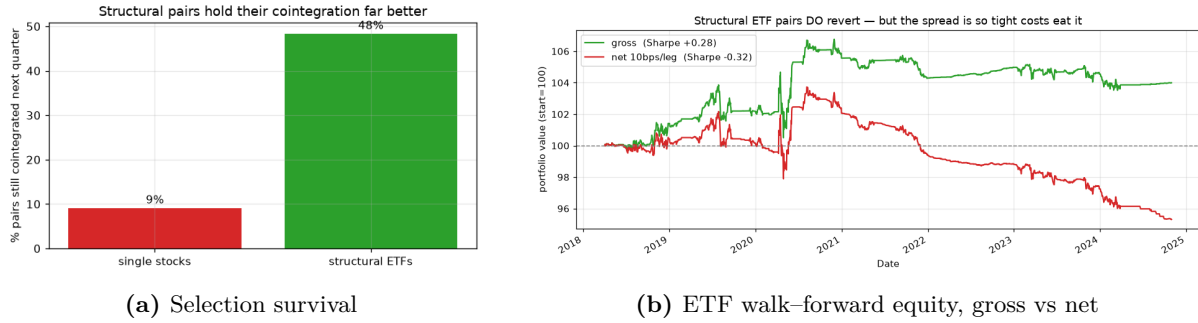


Figure 5: Structural pairs hold their cointegration far better (left) and mean-revert gross of costs — but tight spreads mean realistic frictions consume the edge (right).

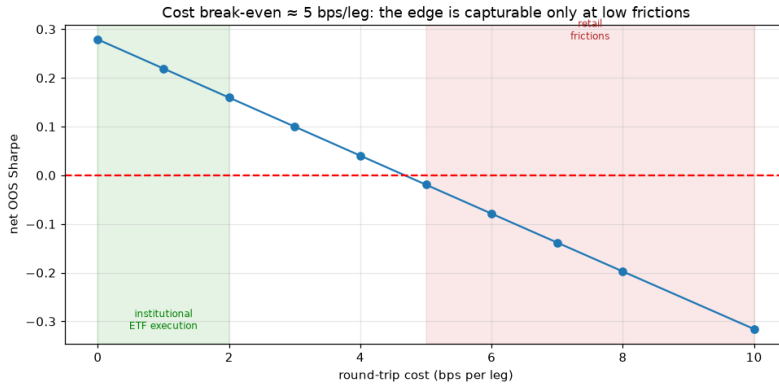


Figure 6: Net out-of-sample Sharpe versus round-trip cost. The edge crosses zero near 5 bps per leg, separating institutional (green) from retail (red) execution regimes.

Why the naive numbers were illusions. The inflated original arose from three compounding effects: a hedge ratio and z-score that used future data, an accounting convention that hid open-position losses, and a pair chosen with knowledge of the full sample. Removing each in turn moved the headline from “spectacular” to “honest,” and the honest number is approximately zero in liquid US equities. This pattern — triple-digit backtested returns accompanied by near-zero risk-adjusted performance once evaluated properly — is common in student and competition work and is precisely what rigorous out-of-sample discipline is designed to expose.

Consistency with the literature. Classical pairs-trading profitability decayed sharply after the early 2000s as the strategy became crowded. Finding no robust net edge in liquid large caps for 2022–2024, and only a thin cost-sensitive edge in structural ETF pairs, is the economically expected outcome rather than a disappointment.

7. Conclusion and Future Work

We rebuilt a naive pairs-trading study into a causal, sector-aware, FDR-controlled, out-of-sample pipeline; found and explained a genuine negative result (universal deintegration); and acted on that diagnosis with an adaptive walk-forward engine and a structural universe. The honest, quantified conclusion is that a real but thin mean-reversion edge exists in structurally cointegrated pairs and is capturable only below ≈ 5 bps per leg. The deliverable is the methodology and its defensibility, not an inflated performance figure.

Future work.

- **Lower the cost floor, not the signal:** trade as a liquidity *provider* (posting limit orders to earn the spread) so the 5 bps wall becomes a tailwind.

Table 4: Full metric suite: Kalman, structural ETFs, 1 bp per leg.

Metric	Value	Metric	Value
Total return	4.96%	Max drawdown	−1.75%
Annualised return	0.74%	% positive days	19.1%
Annualised vol	1.64%	Bootstrap Sharpe med.	0.47
Sharpe	0.46	Pr(Sharpe > 0)	94.2%
Sortino	0.41	Calmar	0.42

- **Breadth:** hundreds of structural pairs — sector ETFs, ADR/ordinary, futures calendar spreads — to diversify many thin edges.
- **Horizon:** an intraday implementation where reversion is faster and spreads are wider relative to the mean–reversion amplitude.
- **Factor neutralisation:** trade the residual after removing common factor exposure so the spread is purely idiosyncratic.

References

- [1] R. F. Engle and C. W. J. Granger, “Co–integration and error correction: Representation, estimation, and testing,” *Econometrica*, vol. 55, pp. 251–276, 1987.
- [2] Y. Benjamini and Y. Hochberg, “Controlling the false discovery rate,” *J. R. Stat. Soc. B*, vol. 57, pp. 289–300, 1995.
- [3] R. E. Kalman, “A new approach to linear filtering and prediction problems,” *J. Basic Eng.*, vol. 82, pp. 35–45, 1960.
- [4] E. Gatev, W. N. Goetzmann, and K. G. Rouwenhorst, “Pairs trading: Performance of a relative–value arbitrage rule,” *Rev. Financ. Stud.*, vol. 19, pp. 797–827, 2006.
- [5] B. Do and R. Faff, “Does simple pairs trading still work?” *Financial Analysts Journal*, vol. 66, pp. 83–95, 2010.
- [6] E. P. Chan, *Algorithmic Trading: Winning Strategies and Their Rationale*. John Wiley & Sons, 2013.
- [7] G. Vidyamurthy, *Pairs Trading: Quantitative Methods and Analysis*. John Wiley & Sons, 2004.